

Literature-Implied Answer to the Near-Tight Feichtinger Partition Exponent

Source: Bownik–Casazza–Marcus–Speegle, arXiv:1508.07353.

Support: Ravichandran–Srivastava, arXiv:1706.03737.

Status. `literature_implied_answer`. The supporting paper does not explicitly state that it is answering Remark 4.9 of arXiv:1508.07353; the answer follows by combining the source paper’s fixed-condition-number Riesz partition theorem with the later asymptotically optimal paving theorem.

Source question. In Section 4 of arXiv:1508.07353, Theorem 4.8 proves that if $(\phi_i)_{i \in I}$ is a unit-norm Bessel sequence with Bessel bound B , then for every $\varepsilon > 0$ it can be partitioned into $O(B\varepsilon^{-4})$ Riesz sequences with bounds $1 - \varepsilon$ and $1 + \varepsilon$. The following Remark 4.9 says that it is open whether this ε -dependence is optimal, notes that the linear dependence on B is optimal, and records a lower bound $\Omega(\varepsilon^{-2})$ as $\varepsilon \rightarrow 0$ from Casazza–Edidin–Kalra–Paulsen.

Supporting theorem. Ravichandran–Srivastava, arXiv:1706.03737, Theorem 1 proves that for zero-diagonal Hermitian contractions $A^{(1)}, \dots, A^{(k)}$ and $\varepsilon > 0$, there is a common paving into at most $18k\varepsilon^{-2}$ blocks with

$$\lambda_{\max}(P_X A^{(j)} P_X) < \varepsilon$$

for every block X and every j . Applying this to $(A, -A)$ gives two-sided paving of every zero-diagonal Hermitian contraction into $O(\varepsilon^{-2})$ blocks. Their abstract also records the consequence that zero-diagonal contractions are $(O(\varepsilon^{-2}), \varepsilon)$ paved.

Literature-implied theorem. Let $(\phi_i)_{i \in I}$ be a unit-norm Bessel sequence in a separable Hilbert space with Bessel bound B . For every $0 < \varepsilon < 1$ there is a partition of I into at most $CB\varepsilon^{-2}$ sets, where C is universal, such that each subfamily is a Riesz sequence with bounds $1 - \varepsilon$ and $1 + \varepsilon$.

Proof. It is enough to discuss finite families; the infinite separable case follows from the same pinball/compactness passage used in Section 4 of the source paper.

By Theorem 4.7 of arXiv:1508.07353, the unit-norm Bessel sequence can first be partitioned into at most $C_0 B$ Riesz sequences with fixed universal Riesz bounds

$$a \sum |c_i|^2 \leq \left\| \sum c_i \phi_i \right\|^2 \leq b \sum |c_i|^2,$$

where $0 < a \leq 1 \leq b < \infty$ are absolute constants.

Fix one piece and let $G = (\langle \phi_j, \phi_i \rangle)_{i,j}$ be its Gram matrix on the corresponding coefficient space. Since the vectors have norm one, G has diagonal entries equal to 1. The Riesz bounds say that the spectrum of G is contained in $[a, b]$. Hence

$$A := G - I$$

is zero diagonal and Hermitian, and $\|A\| \leq K := \max\{1 - a, b - 1\}$, where K is universal.

Apply Ravichandran–Srivastava two-sided paving to A/K with tolerance ε/K ; equivalently apply their Theorem 1 to the two contractions A/K and $-A/K$. This refines the fixed Riesz piece into at most $C_1 K^2 \varepsilon^{-2} = O(\varepsilon^{-2})$ subpieces such that for every refined block J ,

$$\|P_J A P_J\| \leq \varepsilon.$$

Therefore the corresponding compressed Gram matrix satisfies

$$(1 - \varepsilon)I_J \leq P_J G P_J \leq (1 + \varepsilon)I_J,$$

which is exactly the statement that $(\phi_i)_{i \in J}$ is a Riesz sequence with bounds $1 - \varepsilon$ and $1 + \varepsilon$.

Multiplying the initial $O(B)$ pieces by the $O(\varepsilon^{-2})$ refinement inside each piece gives the claimed $O(B\varepsilon^{-2})$ partition size. $\square$

Scope and optimality. This answers the exponent gap in Remark 4.9 up to universal constants. The supporting result supplies the missing ε^{-2} paving exponent. The source paper already notes that the linear dependence on B is optimal, and that $\Omega(\varepsilon^{-2})$ blocks are necessary as $\varepsilon \rightarrow 0$.

References.

- M. Bownik, P. G. Casazza, A. W. Marcus, D. Speegle, *Improved bounds in Weaver and Feichtinger Conjectures*, arXiv:1508.07353.
- M. Ravichandran, N. Srivastava, *Asymptotically Optimal Multi-Paving*, arXiv:1706.03737.
- P. Casazza, D. Edidin, D. Kalra, V. Paulsen, *Projections and the Kadison–Singer problem*, Oper. Matrices 1 (2007), 391–408.